

PROBLEM STATEMENT

Quotient of Power & Linear Functions

Find the derivative with respect to x of the function:

$$y = \frac{(4x - 3)^5}{2x - 1}$$

STEP 1 প্রদত্ত ভগ্নাংশীয় ফাংশনটি উল্লেখ করি (Given Rational Function)

$$y = \frac{(4x - 3)^5}{2x - 1}$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} \left[\frac{(4x - 3)^5}{2x - 1} \right]$$

STEP 3 ভাগবিধি বা Quotient Rule প্রয়োগ করি $\left(\frac{d}{dx} \left[\frac{u}{v} \right] = \frac{vu' - uv'}{v^2} \right)$

$$\frac{dy}{dx} = \frac{(2x - 1) \cdot \frac{d}{dx} [(4x - 3)^5] - (4x - 3)^5 \cdot \frac{d}{dx} (2x - 1)}{(2x - 1)^2}$$

STEP 4 লব ও হরের অন্তরক নির্ণয় করি (Evaluating Individual Derivatives)

$$\frac{d}{dx} [(4x - 3)^5] = 5(4x - 3)^4 \cdot 4 = 20(4x - 3)^4$$

$$\frac{d}{dx} (2x - 1) = 2$$

STEP 5 মান বসিয়ে সাধারণ উৎপাদক $2(4x - 3)^4$ কমন নিয়ে লব সরল করি (Factoring Common Term & Simplifying)

$$\frac{dy}{dx} = \frac{2(4x - 3)^4 [10(2x - 1) - (4x - 3)]}{(2x - 1)^2} = \frac{2(4x - 3)^4 (20x - 10 - 4x + 3)}{(2x - 1)^2} = \frac{2(16x - 7)(4x - 3)^4}{(2x - 1)^2}$$

FINAL ANSWER

$$\frac{dy}{dx} = \frac{2(16x - 7)(4x - 3)^4}{(2x - 1)^2}$$

✦ BOX 1: FORMULATION & QUOTIENT-POWER RULES

- **Quotient Formula:** $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$.
- For $u = (4x - 3)^5$, $u' = 20(4x - 3)^4$.
- For $v = 2x - 1$, $v' = 2$.
- Factoring $2(4x - 3)^4$ leaves: $10(2x - 1) - (4x - 3) = 20x - 10 - 4x + 3 = 16x - 7$.
- Result: $\frac{2(16x-7)(4x-3)^4}{(2x-1)^2}$.

⚠ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Sign Error inside Bracket:** Subtracting $-(4x - 3)$ gives $-4x + 3$. A frequent student mistake is writing $-4x - 3$, giving an erroneous $16x - 13$.
- **Missing the Factor of 4:** Differentiating $(4x - 3)^5$ as $5(4x - 3)^4$ instead of $20(4x - 3)^4$.

🌐 BOX 3: REAL DOMAIN & ASYMPTOTE ANALYSIS

- **Domain of Definition:** The denominator cannot be zero: $2x - 1 \neq 0 \iff x \neq \frac{1}{2}$.

$$\text{Domain} = \mathbb{R} \setminus \left\{ \frac{1}{2} \right\}$$

- **Stationary Points:** $\frac{dy}{dx} = 0$ when:
 1. $4x - 3 = 0 \implies x = \frac{3}{4} = 0.75$ (inflection/flat point).
 2. $16x - 7 = 0 \implies x = \frac{7}{16} = 0.4375$ (local maximum on $x < 0.5$).

💡 BOX 4: LOGARITHMIC DIFFERENTIATION ALTERNATIVE

- Taking natural logarithm:

$$\ln y = 5 \ln(4x - 3) - \ln(2x - 1)$$

- Differentiating:

$$\frac{1}{y} \frac{dy}{dx} = \frac{20}{4x - 3} - \frac{2}{2x - 1} = \frac{20(2x - 1) - 2(4x - 3)}{(4x - 3)(2x - 1)} = \frac{2(16x - 7)}{(4x - 3)(2x - 1)}$$

- Multiplying by $y = \frac{(4x-3)^5}{2x-1}$:

$$\frac{dy}{dx} = \frac{2(16x - 7)(4x - 3)^4}{(2x - 1)^2}$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 4 / 5 MARKS)

- **1.0 Mark:** Correct statement and setup of Quotient Rule.
- **1.0 Mark:** Accurate chain rule differentiation of numerator $20(4x - 3)^4$.
- **1.0 Mark:** Factoring out $2(4x - 3)^4$.
- **1.0 Mark:** Accurate linear algebra simplification to $\frac{2(16x-7)(4x-3)^4}{(2x-1)^2}$.